



C - Mini Project

Under The Guidance of Our Respected :

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Number Guessing Game

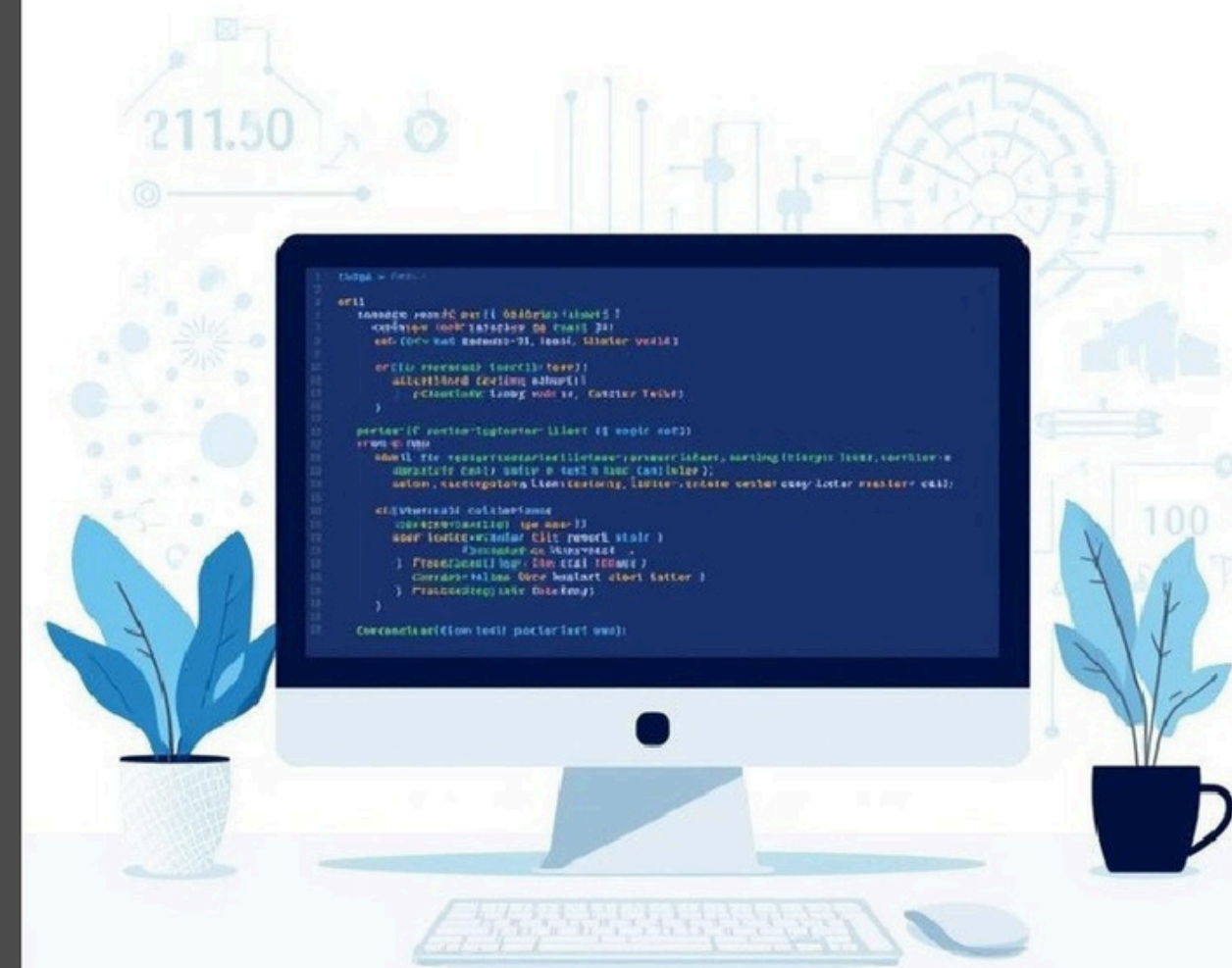
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Introduction

UNDERSTANDING THE NUMBER GUESSING GAME

In a number guessing game, players embark on the challenge of predicting a randomly selected number, which serves as a fun and engaging way to deepen their grasp of C programming fundamentals and logical reasoning.

The C-based Number Guessing Game emerges as an interactive console application specifically crafted to examine player's logic and deductive skills. By integrating elements such as random number generation alongside conditional loops, it creates a dynamic and ever-changing experience. Here, the system daringly invites players to discover a concealed number. This project not only highlights essential programming principles like user input management but also demonstrates real-time algorithmic responses, offering an immersive and educational journey.

Game Rules

1

The program randomly selects a number within a defined range, ensuring each game session is unique and unpredictable for players.

2

GUESSING PROCESS

Players enter their guesses in an attempt to identify the computer's chosen number, promoting engagement and critical thinking throughout the game.

3

FEEDBACK SYSTEM

The program provides immediate feedback on guesses, indicating whether they are "Too High," "Too Low," or "Correct," guiding players toward the solution.

Relevant C Language Concepts

► VARIABLES

In C, variables are fundamental for storing data. They allow the program to hold values like numbers for the guessing game.

► INPUT/OUTPUT

Input/output functions such as printf and scanf facilitate user interaction, enabling players to input their guesses and receive feedback.

► CONDITIONAL STATEMENTS

Conditional statements, primarily if-else structures, control the game's flow, determining if guesses are too high, too low, or correct.

► Loops

A do...while loop in C is a post-tested loop that executes the loop body at least once, regardless of whether the condition is true or false initially. The condition is checked after executing the loop body, making it different from a while loop.

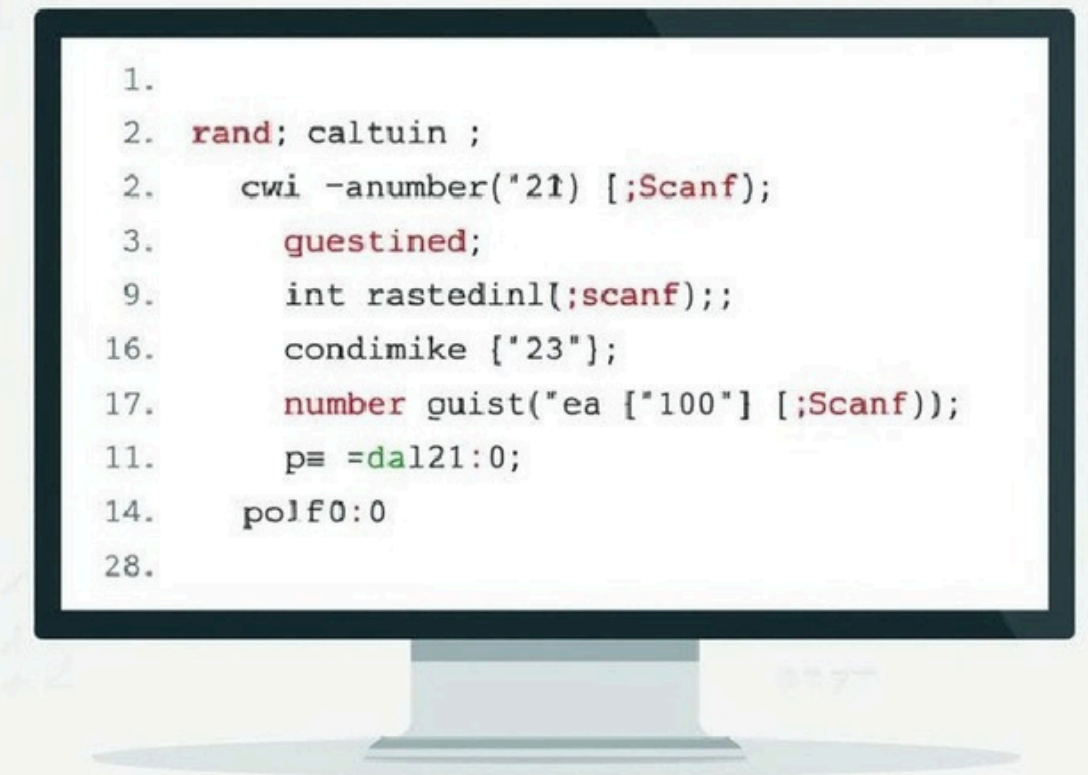
► Libraries Used

- 1 .<stdio.h>
- 2 .<stdlib.h>
- 3 .<time.h>

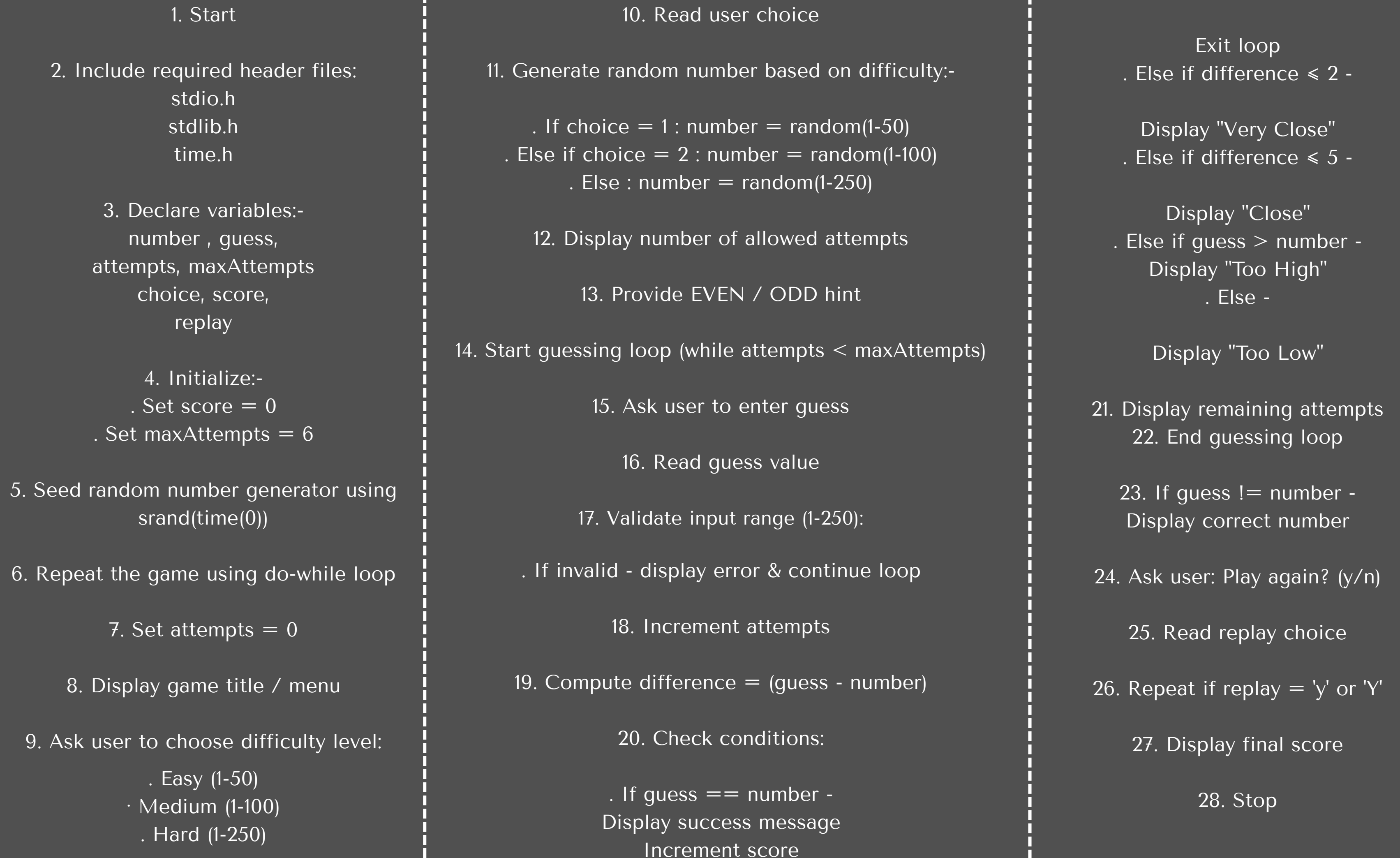
Problem Statement – Number Guessing Game in C

OVERVIEW OF MAIN LOGIC

The objective of this project is to develop a Number Guessing Game using the C programming language. In this game, the computer randomly generates a number within a specified range, and the user is required to guess that number. The program provides hints such as whether the guessed number is even or odd, too high or too low. The game continues until the user correctly guesses the number. The program also counts the number of attempts taken by the user to guess the correct number. This project helps in understanding basic C programming concepts such as random number generation, loops, conditional statements, and user input/output operations.



Algorithm : Number Guessing Game in C.



Flowchart of Number Guessing Game Logic



OUTPUT

```
=====
==== NUMBER GUESSING GAME ====
```

```
=====
Choose Difficulty Level:
```

- 1. Easy (1 - 50)
- 2. Medium (1 - 100)
- 3. Hard (1 - 250)

```
Enter choice: 1
```

```
Guess the number!
```

```
You have 6 attempts
```

```
Hint: The number is ODD
```

```
Enter your guess: 11
```

```
Too Low!
```

```
Attempts left: 5
```

```
Enter your guess: 23
```

```
Correct! You guessed the number!
```

```
Attempts used: 2
```

```
Score: 2
```

```
Play again? (y/n): Y
```

Summary of Key Concepts

GAME CONCEPT

The number guessing game engages players by challenging them to **guess a hidden number**, enhancing their problem-solving and analytical skills.

C PROGRAMMING

This project reinforces fundamental C programming concepts, including **variables**, **control structures**, and functions essential for developing interactive applications.

VISUAL FLOW

Understanding the game logic through a **flowchart** helps visualize the program's structure and decision-making process, ensuring clarity and effectiveness in coding.

.THANKYOU!

Your feedback is appreciated!!!

Please Feel Free to Ask Any Question...